



Interactive EDA Dashboard

Data Science (24AD3107) | Academic Year 2026–2027

Team Members

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Project Guide

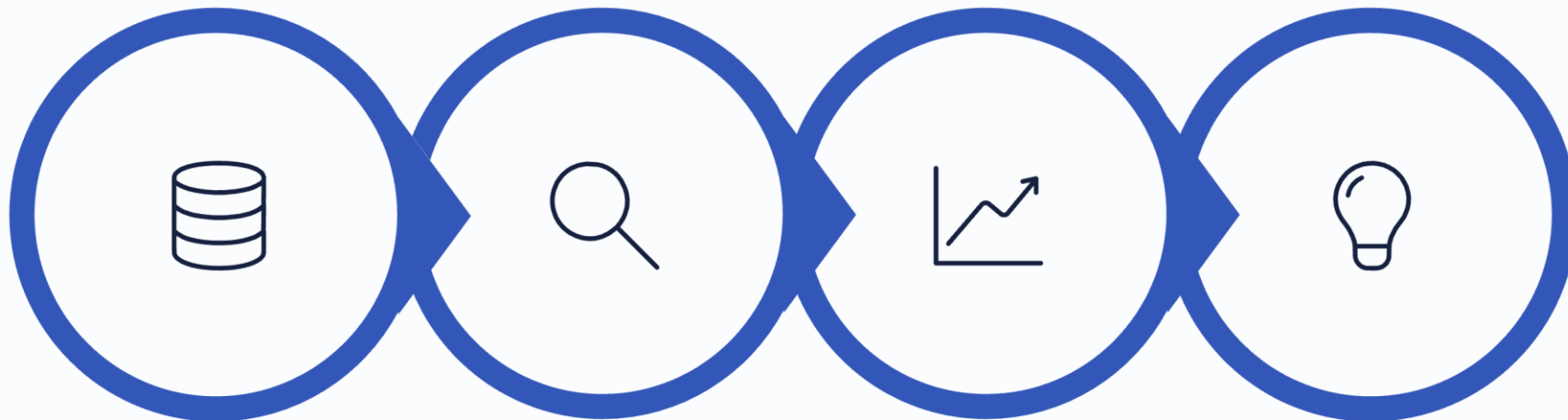
Dr. N. Sunanda



FOUNDATION

What is Exploratory Data Analysis?

- EDA is the **first critical step** in understanding raw datasets before deeper analysis
- Raw datasets are often large, unstructured, and difficult to interpret directly
- Manual exploration requires multiple steps: inspection, cleaning, visualization, and interpretation
- Interactive tools streamline this process and reveal **hidden patterns**



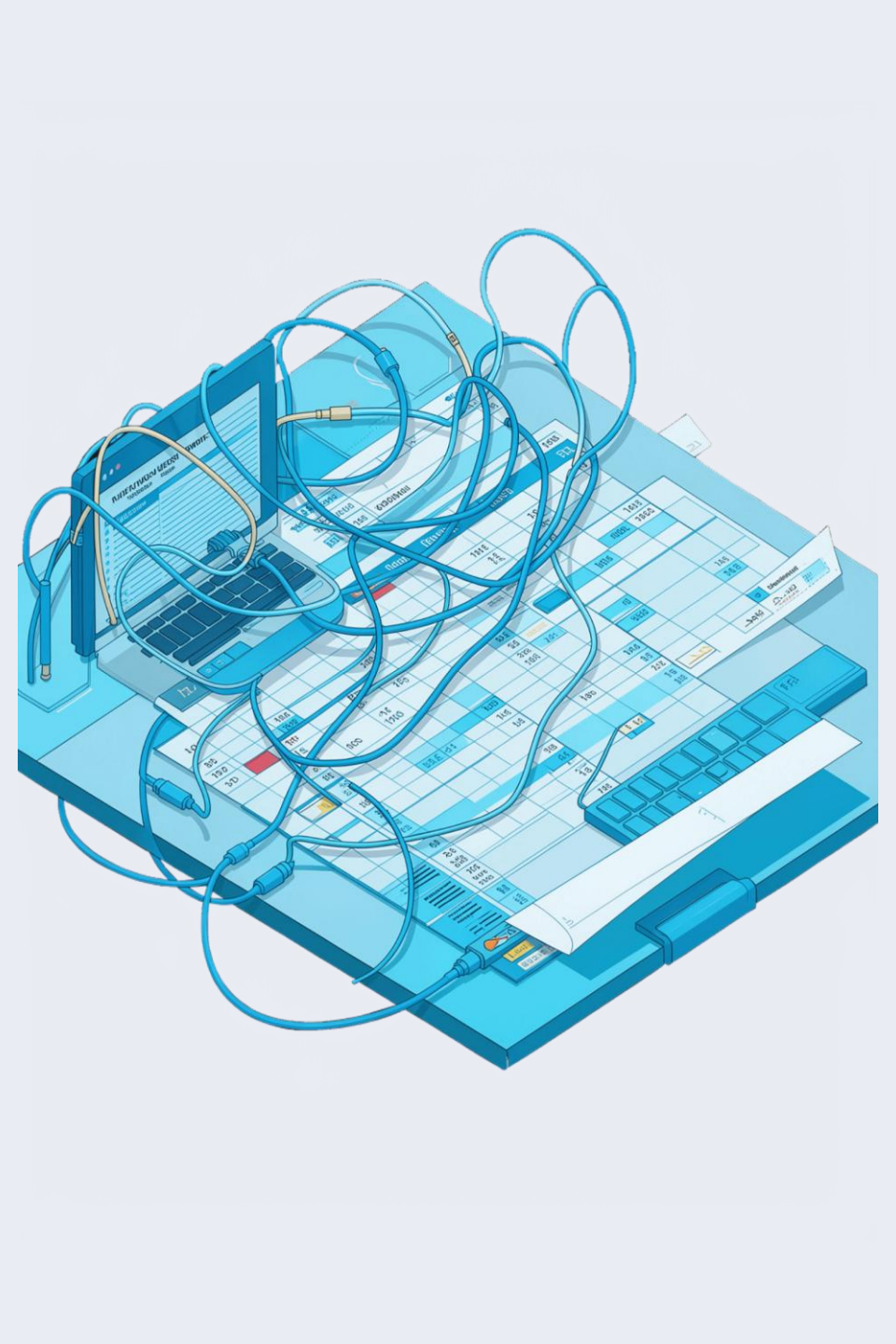
Problem Statement & Objectives

Challenges

- Raw datasets are difficult to understand; manual analysis is **time-consuming and error-prone**
- Identifying missing values, duplicates, and inconsistencies requires significant effort
- Creating visualizations manually requires **coding expertise**

Our Goals

- Simplify EDA and make it **accessible to all skill levels**
- Automatically detect data-quality issues and generate visualizations
- Enable interactive exploration and pattern discovery in **one unified platform**



OUR SOLUTION

Proposed Solution

A unified, user-friendly platform automating the entire exploratory data analysis workflow:



Dataset Upload

Simple file upload interface for instant onboarding



Auto Preprocessing

Data inspection and quality checks, automatically



Statistical Summary

Descriptive statistics and data profiling at a glance



Interactive Visualizations

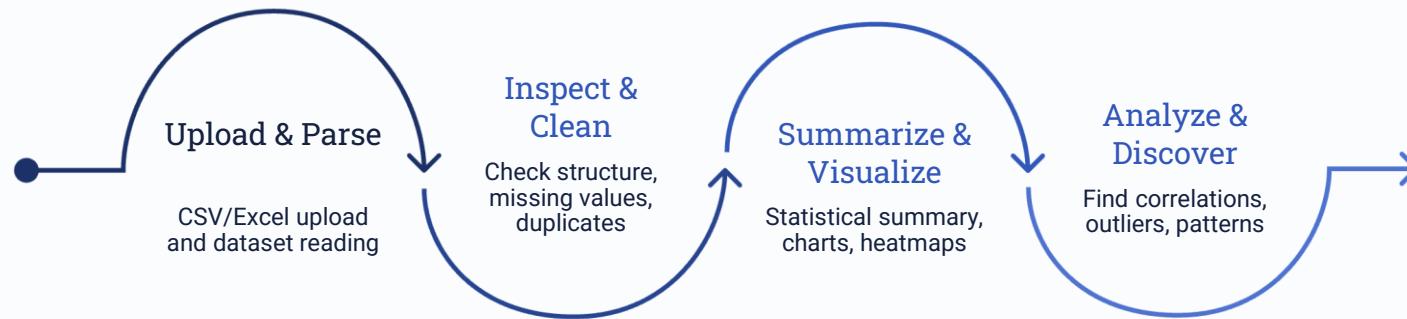
Multiple chart types generated automatically



Insight Generation

Patterns, relationships, and anomalies identified instantly

System Workflow & Methodology



01

Upload Dataset

CSV, Excel, or similar format

02

Inspect Structure

Rows, columns, and data types

03

Check Data Quality

Missing values, duplicates, inconsistencies

04

Statistical Summary

Mean, median, std dev, quartiles

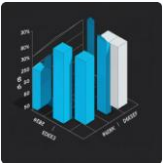
05

Visualize & Analyze

Charts, correlations, outlier detection

FEATURES

Key Features & Visualizations



Bar Chart & Histogram

Compare categorical values and visualize **distribution of numerical data** with clarity



Box Plot & Scatter Plot

Identify outliers and reveal relationships between variables through interactive plots



Correlation Heatmap

Visualize **feature dependencies** across all variables in a single, comprehensive view

Dashboard UI Design & Technology Stack

Dashboard Layout

Top Panel

Upload interface and file information

Left Sidebar

Navigation controls and filters

Main Area

Data preview and interactive visualizations

Right Panel

Statistics summary and correlation heatmap

Bottom Section

Auto-generated insights and findings

Technology Stack

Python

Core programming language

Pandas & NumPy

Data manipulation and numerical operations

Matplotlib & Seaborn

Statistical visualization

Streamlit

Interactive web-based dashboard interface

Benefits & Expected Results



Key Benefits

- Reduces **hours of manual analysis to minutes**
- Enables non-technical users to perform professional-grade EDA
- Automatically identifies data-quality issues
- Reveals **hidden patterns and relationships** effortlessly

Expected Insights Delivered

- Dataset structure understanding and profiling
- Missing value and duplicate identification
- Distribution visualization and outlier detection
- Correlation discovery and **actionable pattern recognition**

LOOKING AHEAD

Future Scope & Enhancements



Expanded Data Sources

JSON, SQL databases, and live API integration



ML-Assisted Analysis

Automated anomaly detection, clustering, and hypothesis testing



Export & Reporting

Exportable PDF, PNG, HTML reports and real-time data streaming

Advanced filtering, interactive drill-down, and automated insight generation are also planned to further elevate the platform's analytical depth.

Conclusion

The Interactive EDA Dashboard transforms exploratory data analysis from a **time-consuming, manual process** into an efficient, intuitive, and accessible workflow.

By automating data inspection, quality checks, and visualization generation, it empowers users to uncover patterns, identify data-quality issues, and extract meaningful insights — all within a **single, unified platform**.

Thank You

